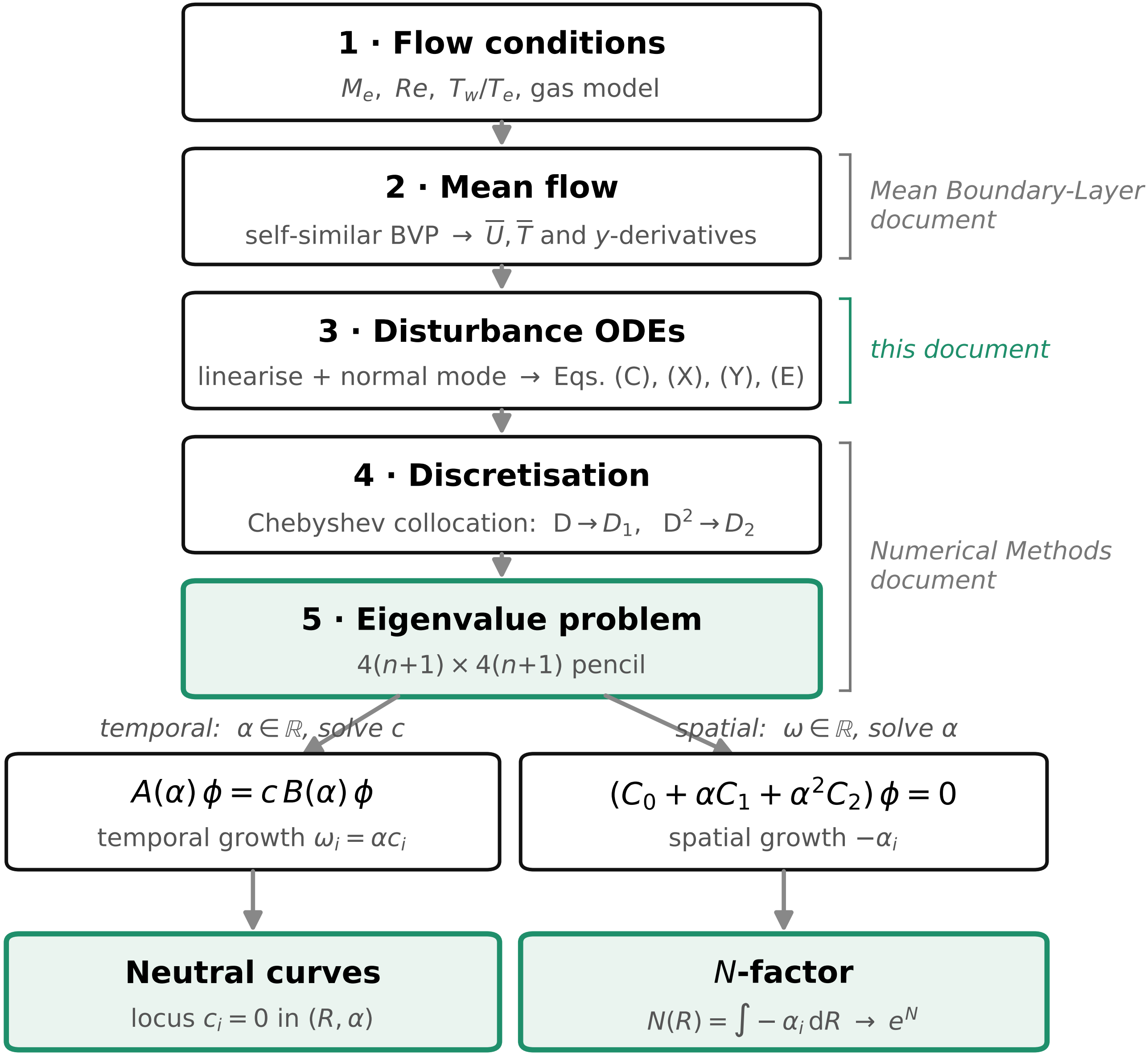
1 · Flow conditions
M_e, Re, T_w/T_e, gas model
2 · Mean flow
self-similar BVP → $\overline{U}, \overline{T}$ and y-derivatives
Mean Boundary-Layer document
3 · Disturbance ODEs
linearise + normal mode → Eqs. (C), (X), (Y), (E)
this document
4 · Discretisation
Chebyshev collocation: $\mathrm{D} \to D_1$, $\mathrm{D}^2 \to D_2$
Numerical Methods document
5 · Eigenvalue problem
$4(n+1) \times 4(n+1)$ pencil
temporal: $\alpha \in \mathbb{R}$, solve c
spatial: $\omega \in \mathbb{R}$, solve α
$A(\alpha)\,\phi = c\,B(\alpha)\,\phi$
temporal growth $\omega_i = \alpha c_i$
$(C_0 + \alpha C_1 + \alpha^2 C_2)\,\phi = 0$
spatial growth $-\alpha_i$
Neutral curves
locus $c_i = 0$ in (R, α)
N-factor
$N(R) = \int -\alpha_i\,\mathrm{d}R \ \to\ e^N$